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FEEDING PREFERENCE OF SHORT-TAILED MOLE RAT, *NESOKIA INDICA*, FOR SOME COMMERCIAL VARIETIES OF SUGARCANE UNDER CONTROLLED CONDITIONS

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ABSTRACT

Laboratory studies were conducted to explore the potential of preference of short-tailed mole rats (*Nesokia indica*) for the four commercial varieties of sugarcane, namely; BL-4, Gulabi-95, Triton and Thatta-10. Results revealed that BL-4 was the most preferred variety; Sugarcane consumption was 55.86 ± 1.41 g as compared to rice consumption 24.46 ± 1.05 g/kg body weight by male rats; whereas sugarcane consumption was 66.86 ± 2.60 g/kg as compared to rice consumption 21.75 ± 1.45 g/kg body weight by female rats. Thatta-10 was found the least preferred variety by showing sugarcane consumption as 23.83 ± 1.04 g/kg as compared to rice consumption 28.62 ± 0.44 g/kg body weight by male rats; whereas sugarcane consumption was 26.33 ± 1.38 g/kg as compared to rice consumption 22.55 ± 1.71 g/kg body weight by female rats. For the variety Triton, sugarcane consumption was 34.25 ± 1.88 g/kg as compared to rice consumption 11.10 ± 0.65 g/kg body weight by male rats; whereas sugarcane consumption was 42.43 ± 1.04 g/kg as compared to rice consumption 24.59 ± 1.33 g/kg body weight by female rats. For the variety Gulabi-95, sugarcane consumption was 43.91 ± 1.75 g/kg as compared to rice consumption 18.07 ± 1.17 g/kg body weight by male rats; whereas sugarcane consumption was 42.51 ± 1.30 g/kg as compared to rice consumption 19.99 ± 1.52 g/kg body weight by female rats.

Key Words: Short-tailed mole rats (*Nesokia indica*), commercial varieties, preference, consumption and g/kg body weight.

INTRODUCTION

Sugarcane (*Saccharum officinarum*) is a widespread crop, growing best in the zone between tropics and sub-tropics. Next to cotton, sugarcane is Pakistan's most important cash crop. It is grown primarily in the making of sugar and sugar-related products. Sugarcane also produces essential items for industries, like chipboard, paper etc. Sugarcane is high value cash crop. Its production accounts for 2.9 percent in agriculture's value

addition and 0.5 percent of overall GDP. During 2018-19, sugarcane crop production was lower by 19.4 percent (to 67.174 million tonnes) compared to 83.333 million tonnes achieved in last year. It was cultivated on an area of 1,102 hectares during 2018-19 (against last year's area of 1,343 hectares) showing a decrease of 17.9 %, producing 60.96 million tons of sugarcane (against 62.05 million tons last year), (Pakistan Economic

Survey 2018-19). Sugarcane is a profitable crop for growers as long as growing problems do not overtake the fields. One of the major problems is invasion of the rodent pests. Sugarcane is a long duration crop and thus provides an ideal condition for a complex of rodent population to subsist and cause extensive damage through eating into internodes and roots resulting infestation with microorganisms, causing the susceptibility for diseases, insect deterioration,

and reduction of weight and sugar recovery. The damage becomes more prominent in lodged sugarcane. Sugarcane crop is heavily damaged by a complex of rodent species including *Nesokia indica*, *Bandicota bengalensis*, *Millardia melitana* and *Mus booduga* (Hussain *et al*, 1975; Smiet *et al*, 1978, 1980; Beg *et al*, 1979 & 1980; Khan, 1982; Hampson, 1984; Roberts, 1997; Mishkat *et al*, 2006). Economic losses by rodents to sugarcane in Pakistan are documented by some scientists, such as Beg *et al*, (1979) described 11% damage to sugarcane in three districts of central Punjab. Fulk *et al*, (1980) assessed 7.2 and 4.4% damaged canes and reduced sugar production by an average of 10.7% and 7.7% in four districts of Punjab and Sindh during 1978 and 1979, respectively. Mishkat *et al*, (2006) estimated 10.77% reduction in sugar recovery because of rat damage in sugarcane growing areas of Mandi Bahauddin, Phalia, Bhalwal and Khushab. Smiet *et al*. (1980) estimated 24% damage to canes in Thatta district of lower Sindh. Khan (1990) calculated economic losses of Rs.140.6 million annually based on 1986-87 production and support prices. He recommended the usage of 0.005% brodifacoum wax blocks (Klerat) to many workers in cane fields. In a rat damage survey by Fulk *et al*, 1980 at Fauji Sugar Mill, Khoski, Badin, 10.7% loss to sugar

production was recorded. In another study conducted in sugarcane growing areas of Army Welfare Sugar Mills, Badin district, 2% Zinc Phosphide baiting decreased 74% rodent population (Khan & Munir, 2006). Many scientists of Vertebrate Pest Control Institute conducted extensive surveys, laboratory and field trials related to rodent management and developed models, packages for the growers (Smythe, 1977; Khan, 1977; Smythe & Khan, 1980; Fulk *et al*, 1978 & 1980; Smiet *et al*, 1980; Shafi & Khan, 1983; Khokhar & Rizvi, 1999; Pervez *et al*, 1999 & 2005; Tariq *et al*, 2009).

MATERIAL AND METHODS

Laboratory studies were carried out to know the preference for sugarcane by the rats as well as the difference of sugarcane consumption among the four commercial varieties of sugarcane, namely, BL-4, Gulabi-95, Triton and Thatta-10 (Figure-1 to Figure-5). Short-tailed mole rats, *Nesokia indica*, were collected through digging of burrows found in sugarcane fields in Thatta district, lower Sindh, Pakistan. Twenty rats (ten male and ten female) were sexed, weighed and caged individually for acclimation in laboratory for 15 days and were fed on poultry feed during acclimation

period and between the trials. Water was provided *ad libitum*. All the four varieties of sugarcane were harvested freshly from National Sugar and Tropical Horticulture Research Institute, Pakistan Agricultural Research Council, Thatta, lower Sindh. Stalk sections were cut from the bottom of mature canes and the pre-weighed sections of all four varieties (individually) of sugarcane stalk were kept in all cages of the rats. Twenty grams rice were provided to each rat as an alternative food source occurring naturally (Martin *et al*, 2007). Trials were carried out in animal rooms (at SARC, PARC premises), away from human activity and rats were kept visually isolated from each other (Figures, 1-5). All the sugar cane stalks and the left-over rice were weighed again after 24 hours. The sugarcane and rice consumption by each rat were recorded for five consecutive days. Comparison of sugarcane intake (g/kg body weight) among the four sugar cane varieties and rice intake (g/kg body weight) by the rats was carried out to know the preference.

RESULTS AND DISCUSSION

Short-tailed mole rat, *Nesokia indica*, inflicts significant damage to sugarcane crop; however, the damage inflicted to different varieties of sugarcane by the specie is unknown. The feeding trial experiment was carried out to know the difference in the

grams of sugarcane consumed for the four commercial varieties of sugarcane, namely, Gulabi-95, Triton, BL-4 and Thatta-10 by the specie. The sugarcane and rice consumption by each rat were recorded for five days. As per results (Table-1), BL-4 was found the most preferred variety. Sugarcane consumption was 55.86 ± 1.41 g/kg body as compared to rice consumption 24.46 ± 1.05 g/kg body weight by male rats; whereas sugarcane consumption was 66.86 ± 2.60 g/kg as compared to rice consumption 21.75 ± 1.45 g/kg body weight by female rats. The variety may be preferred due to its softness and high sucrose contents. Thatta-10 was found the least preferred variety by showing sugarcane consumption as 23.83 ± 1.04 g/kg as compared to rice consumption 28.62 ± 0.44 g/kg body weight by male rats; whereas sugarcane consumption was 26.33 ± 1.38 g/kg as compared to rice consumption 28.62 ± 0.44 g/kg body weight by female rats. For the variety Triton, sugarcane consumption was 34.25 ± 1.88 g/kg as compared to rice consumption 11.10 ± 0.65 g/kg body weight by male rats; whereas sugarcane consumption was 42.43 ± 1.04 g/kg as compared to rice consumption 24.59 ± 1.33 g/kg body weight

by female rats. For the variety Gulabi-95, sugarcane consumption was 43.91 ± 1.75 g/kg as compared to rice consumption 18.07 ± 1.17 g/kg body weight by male rats; whereas sugarcane consumption was 42.51 ± 1.30 g/kg as compared to rice consumption 19.99 ± 1.52 g/kg body weight by female rats (Table-1, Graph-1 & Graph-2). Overall sugarcane was preferred for all the varieties except Thatta-10 as compared to rice. Thatta-10 proved it the best variety in the contest; although it has the highest sucrose contents (Keerio et al, 2003). However, due to its hardness and thickness it was not preferred by the rats. The male rats preferred BL-4 variety the most and Triton the least in comparison to rice; whereas Thatta-10 variety was not preferred in comparison to rice. The female rats preferred BL-4 the most and Thatta-10 the least in comparison to rice; however for Thatta-10 the difference among the intake of sugarcane and rice was only 3.78 g, which may be due to the high sucrose contents of the variety. The preference of the rats for sugar cane is shown in Graph 1 & 2. The female rats may require the high

sucrose for the reproduction. The rats significantly preferred all the sugar cane varieties in comparison to rice (Table-2). These results are in line with previous findings by Tariq et al (2009).

RECOMMENDATIONS

Sugarcane is a long duration crop and thus provides an ideal condition for a complex of rodent population to subsist and cause extensive damage to the sugarcane, standing or lodged. The rodent control often performed is insufficient, in response to high rodent populations, poorly timed or is after the damage has been done. The growers need a better understanding of the rodent damage as per the variety, rodent species identification and its biology in relation to crop timings. A cost-benefit analysis should be conducted for rodent management. An Integrated Pest Management approach is being practiced to manage rodent's damage in sugarcane crop by Vertebrate Pest Control Institute, Southern-zone Agricultural Research Centre, Pakistan Agricultural Research Council. The package is successfully developed for rodent management after a series of trials and is now ready for adoption by the end-users.

Figure-1: Sections of sugarcane variety BL-4**Figure-2: Sections of sugarcane variety Gulabi-95****Figure-3: Sections of sugarcane variety Triton****Figure-4: Sections of sugarcane variety Thatta-10****Figure-5: Laboratory trials on feeding preference of *Nesokia indica*****Table-1: Consumption (g/kg body weight) of Sugarcane and Rice by short-tailed mole rat, *Nesokia indica***

Sugarcane Variety	Male Rat			Female Rat		
	Weight	Sugarcane	Rice	Weight	Sugarcane	Rice
Triton	154.80 ±3.15	34.25±1.88	11.10±0.65	121.80± 2.56	42.43±1.04	24.59±1.33
BL-4	139.40 ±2.25	55.86±1.41	24.46±1.05	128.00 ±2.88	66.86±2.60	21.75±1.45
Gulabi-95	152.60 ±3.36	43.91±1.75	18.07±1.17	124.40 ±3.25	42.51±1.30	19.99±1.52
Thatta-10	137.40 ±1.81	23.83±1.04	28.62±0.44	128.80 ±2.22	26.33±1.38	22.55±1.71
LSD ^{0.05}	8.14	4.65	2.62	8.25	5.06	4.54

Table-2: ANOVA for the Consumption (g/kg body weight) of Sugarcane and Rice by short-tailed mole rat, *Nesokia indica*

		Sum of Squares	df	Mean Square	F	Sig.
Weigh of male rats (g)	Between Groups	1192.55	3	397.516667	10.7728094	0.00040531
	Within Groups	590.4	16	36.9		
	Total	1782.95	19			
Sugarcane Intake	Between Groups	2800.70474	3	933.568247	77.2895935	9.7701E-10
	Within Groups	193.26136	16	12.078835		
	Total	2993.9661	19			
Rice Intake	Between Groups	882.241735	3	294.080578	76.4842436	1.0563E-09
	Within Groups	61.51972	16	3.8449825		
	Total	943.761455	19			
Weight of female rats (g)	Between Groups	158.95	3	52.9833333	1.39705559	0.27999662
	Within Groups	606.8	16	37.925		
	Total	765.75	19			
Sugarcane Intake	Between Groups	4192.84826	3	1397.61609	97.9032124	1.6545E-10
	Within Groups	228.4078	16	14.2754875		
	Total	4421.25606	19			
Rice Intake	Between Groups	54.4873	3	18.1624333	1.58367924	0.23238196
	Within Groups	183.49608	16	11.468505		
	Total	237.98338	19			

The male rats preferred BL-4 variety the most and Triton the least in comparison to rice; whereas Thatta-10 variety was not preferred in comparison to rice.

The female rats preferred BL-4 the most and Thatta-10 the least in comparison to rice; however for Thatta-10 the difference among the intake of sugarcane and rice was only 3.78 g.

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